Table 1. Individual root traits across T1–T4: 0353

Observation unit	Period	Skeleton length (10^3 px)	Persistent (%)	New (%)	Extension (%)	E / Lprev (%)
0353-1-B3	T1	6.35	—	—	—	—
	T2	16.11	28.64	59.07	12.29	31.19
	T3	16.42	73.64	19.09	7.28	7.42
	T4	7.52	95.09	2.07	2.84	1.30
0353-2-B3	T1	29.25	—	—	—	—
	T2	45.53	58.57	36.59	4.83	7.53
	T3	52.29	83.73	13.95	2.32	2.66
	T4	47.51	96.54	3.18	0.28	0.26
0353-2-LB2†	T1	16.35	—	—	—	—
	T2	26.91	45.62	48.62	5.76	9.48
	T3	27.55	85.39	10.89	3.72	3.81
	T4	24.06	90.88	5.44	3.67	3.21
0353-2-LB3	T1	36.52	—	—	—	—
	T2	46.02	64.75	31.49	3.76	4.74
	T3	45.71	88.70	8.98	2.32	2.30
	T4	38.79	96.84	2.62	0.54	0.46
0353-3-LB2	T1	5.10	—	—	—	—
	T2	4.33	69.26	19.06	11.68	9.90
	T3	4.21	80.16	16.79	3.05	2.97
	T4	5.70	46.45	45.06	8.48	11.49
0353-3-LB3†	T1	17.34	—	—	—	—
	T2	12.07	73.41	19.74	6.85	4.77
	T3	12.38	82.82	10.52	6.66	6.83
	T4	7.13	95.52	2.16	2.32	1.34

Individual sequence estimates are shown, without averaging across materials or adding an SD. Length is reported in 10^3 native registered pixels; physical calibration is not confirmed. For T2, T3 and T4, temporal percentages refer to T1–T2, T2–T3 and T3–T4, respectively. Persistent, New and Extension are current-class length/current total length ×100%. E / Lprev is extension-class length/previous total length ×100%, not a time-normalized growth rate. "—" denotes baseline, missing acquisition or unavailable valid adjacent comparison; real zeros remain 0.00. The 18 observation sequences belong to 8 source tubes and are not 18 independent biological replicates. This linked workbook contains 63 usable length observations and 40 valid adjacent pairs; 6 unobserved periods and 3 unavailable source results remain missing. Existing V35/V66 U-Net model estimates and review flags are retained; these are not V90 measurements.

0353-2-LB2: † Existing registration review flag retained. 0353-3-LB3: † Existing registration review flag retained.

Table 2. Individual root traits across T1–T4: A017

Observation unit	Period	Skeleton length (10^3 px)	Persistent (%)	New (%)	Extension (%)	E / Lprev (%)
A017-1-B2	T1	16.78	—	—	—	—
	T2	23.37	50.50	41.22	8.27	11.52
	T3	—	—	—	—	—
	T4	16.25	—	—	—	—
A017-1-Z2	T1	5.53	—	—	—	—
	T2	7.27	33.81	54.20	11.99	15.76
	T3	—	—	—	—	—
	T4	5.88	—	—	—	—
A017-2-2†	T1	6.56	—	—	—	—
	T2	12.69	31.54	59.74	8.72	16.87
	T3	12.12	59.84	27.46	12.70	12.13
	T4	8.78	59.90	26.05	14.05	10.17
A017-2-3	T1	7.75	—	—	—	—
	T2	6.16	51.40	27.32	21.28	16.91
	T3	12.40	33.42	48.20	18.38	37.03
	T4	3.90	81.44	7.61	10.95	3.44
A017-3-B3†	T1	19.51	—	—	—	—
	T2	21.98	74.98	19.65	5.37	6.05
	T3	14.75	74.04	13.41	12.55	8.43
	T4	10.49	87.41	3.88	8.71	6.19
A017-3-Z2†	T1	22.22	—	—	—	—
	T2	19.78	66.08	24.83	9.09	8.09
	T3	14.94	70.87	14.73	14.40	10.88
	T4	8.79	77.86	10.07	12.07	7.10

Individual sequence estimates are shown, without averaging across materials or adding an SD. Length is reported in 10^3 native registered pixels; physical calibration is not confirmed. For T2, T3 and T4, temporal percentages refer to T1–T2, T2–T3 and T3–T4, respectively. Persistent, New and Extension are current-class length/current total length ×100%. E / Lprev is extension-class length/previous total length ×100%, not a time-normalized growth rate. "—" denotes baseline, missing acquisition or unavailable valid adjacent comparison; real zeros remain 0.00. The 18 observation sequences belong to 8 source tubes and are not 18 independent biological replicates. This linked workbook contains 63 usable length observations and 40 valid adjacent pairs; 6 unobserved periods and 3 unavailable source results remain missing. Existing V35/V66 U-Net model estimates and review flags are retained; these are not V90 measurements.

A017-1-B2: Acquisition missing: T3. Temporal shares unavailable at T4 because the immediately preceding period is unavailable.
A017-1-Z2: Acquisition missing: T3. Temporal shares unavailable at T4 because the immediately preceding period is unavailable.
A017-2-2: † Existing registration review flag retained. A017-3-B3: † Existing registration review flag retained. A017-3-Z2: † Existing registration review flag retained.

Table 3. Individual root traits across T1–T4: Q1

Observation unit	Period	Skeleton length (10^3 px)	Persistent (%)	New (%)	Extension (%)	E / Lprev (%)
Q1-1-B2	T1	6.78	—	—	—	—
	T2	—	—	—	—	—
	T3	5.01	—	—	—	—
	T4	2.94	84.65	7.45	7.90	4.63
Q1-1-Z2†	T1	4.41	—	—	—	—
	T2	10.40	38.86	57.41	3.72	8.78
	T3	7.57	74.42	16.96	8.62	6.28
	T4	2.65	77.06	3.91	19.03	6.66
Q1-1-Z3†	T1	6.74	—	—	—	—
	T2	9.87	83.85	13.94	2.21	3.24
	T3	12.59	45.32	44.44	10.24	13.06
	T4	—	—	—	—	—
Q1-2-B3	T1	—	—	—	—	—
	T2	—	—	—	—	—
	T3	—	—	—	—	—
	T4	78.56	—	—	—	—
Q1-2-Z2†	T1	35.13	—	—	—	—
	T2	—	—	—	—	—
	T3	60.30	—	—	—	—
	T4	61.51	85.45	13.16	1.38	1.41
Q1-2-Z3†	T1	53.04	—	—	—	—
	T2	150.63	34.25	50.14	15.60	44.32
	T3	—	—	—	—	—
	T4	148.72	—	—	—	—

Individual sequence estimates are shown, without averaging across materials or adding an SD. Length is reported in 10^3 native registered pixels; physical calibration is not confirmed. For T2, T3 and T4, temporal percentages refer to T1–T2, T2–T3 and T3–T4, respectively. Persistent, New and Extension are current-class length/current total length ×100%. E / Lprev is extension-class length/previous total length ×100%, not a time-normalized growth rate. "—" denotes baseline, missing acquisition or unavailable valid adjacent comparison; real zeros remain 0.00. The 18 observation sequences belong to 8 source tubes and are not 18 independent biological replicates. This linked workbook contains 63 usable length observations and 40 valid adjacent pairs; 6 unobserved periods and 3 unavailable source results remain missing. Existing V35/V66 U-Net model estimates and review flags are retained; these are not V90 measurements.

Q1-1-B2: Acquisition missing: T2. Temporal shares unavailable at T3 because the immediately preceding period is unavailable. Q1-1-Z2: † Existing registration review flag retained. Its T2–T3 coarse correspondence remains unverified. Q1-1-Z3: Acquisition missing: T4. † Existing registration review flag retained. Q1-2-B3: Source results unavailable in the linked workbook: T1, T2, T3. Temporal shares unavailable at T4 because the immediately preceding period is unavailable. Q1-2-Z2: Acquisition missing: T2. Temporal shares unavailable at T3 because the immediately preceding period is unavailable. † Existing registration review flag retained. Q1-2-Z3: Acquisition missing: T3. Temporal shares unavailable at T4 because the immediately preceding period is unavailable. † Existing registration review flag retained.